HX-200 Hydraulic Pump Service Manual

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Model: HX-200 axial piston pump

1 Introduction

This manual covers inspection, adjustment and repair of the HX-200 axial piston pump used on mobile hydraulic equipment. Read the safety section before starting any work.

1.1 Safety

WARNING: Relieve all hydraulic pressure before disconnecting any line. Fluid escaping under pressure can penetrate the skin and cause serious injury.

CAUTION: Pump surfaces can exceed 80 C during operation. Allow the pump to cool before service.

2 Specifications

The values below apply to pumps with serial numbers from HX2A00001 onward.

Table 1 HX-200 performance specifications

Parameter	Value	Notes
Rated pressure	350 bar (5076 psi)	Continuous duty
Peak pressure	420 bar (6092 psi)	Maximum 10 s
Rated flow	120 L/min	At 1800 rpm
Displacement	67 cm3/rev	Fixed
Relief valve setting	380 bar	Factory set
Dry weight	48 kg	Without fluid
Operating temperature	-20 to 80 C	Fluid temperature
Shaft seal kit	4J-5521	Part number

3 Torque Specifications

Table 2 Fastener torque values

Fastener	Thread	Torque (N.m)	Torque (lb.ft)
Mounting flange bolt	M16x2.0	210	155
Port cover bolt	M10x1.5	49	36
Drain plug	M18x1.5	60	44

Apply thread-locking compound 9S-3263 to the mounting flange bolts before installation. Tighten the bolts in a cross pattern in two stages.

4 Shaft Seal Replacement

Special tools: seal driver 1U-7650 and seal puller 5P-3520.

> **WARNING: Lower all attachments to the ground and relieve hydraulic pressure before removing the pump.**

1. Drain the pump case through the drain plug and discard the fluid.
2. Remove the four port cover bolts and the port cover.
3. Remove the retaining ring from the shaft bore.
4. Use seal puller 5P-3520 to remove the old shaft seal. Do not scratch the shaft.
5. Lubricate the new seal from kit 4J-5521 with clean hydraulic oil.
6. Install the new seal with seal driver 1U-7650 until it is flush with the bore.
7. Install the retaining ring and the port cover. Tighten the port cover bolts to 49 N.m.

After assembly, fill the case with clean fluid and run the pump at low idle for 5 minutes. Check for leaks at the shaft seal.

5 Troubleshooting

Pump noise

Symptom: the pump makes a loud whining noise under load.

Cause: cavitation caused by a clogged suction strainer.

Remedy: clean the suction strainer and check that inlet vacuum does not exceed 0.2 bar.

Low output pressure

Symptom: the system cannot reach rated pressure.

Cause: a worn or contaminated relief valve cartridge.

Remedy: replace relief valve cartridge 1R-0750 and verify the relief setting.

6 Fluid Recommendations

Use ISO VG 46 anti-wear hydraulic oil. Change filter element 1R-0741 every 500 service hours. Change the hydraulic fluid every 2000 service hours or after any pump failure.

7 Hydraulic Circuit

Figure 1 shows the HX-200 in a typical open-loop circuit. Oil is drawn from the tank through the suction strainer, delivered by the pump to the directional control valve and returned to the tank through return filter 1R-0741.

The relief valve protects the circuit and is factory set to 380 bar.

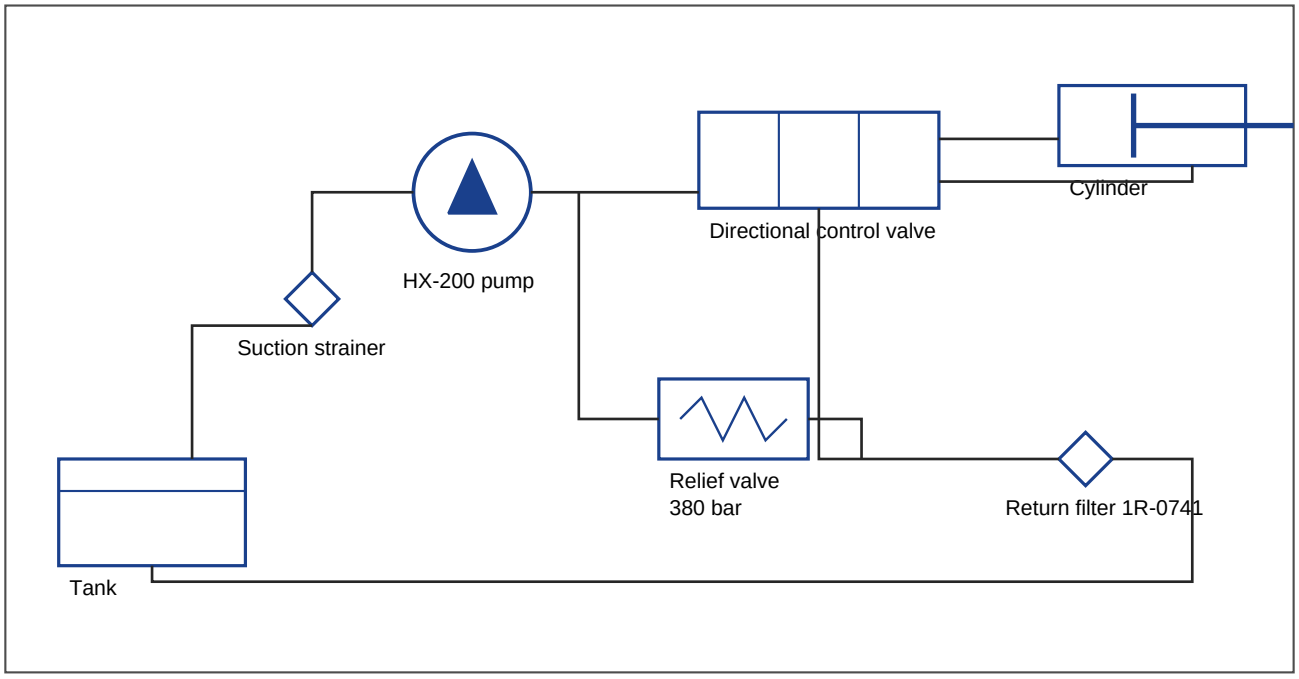

Figure 1 HX-200 hydraulic circuit (open loop)

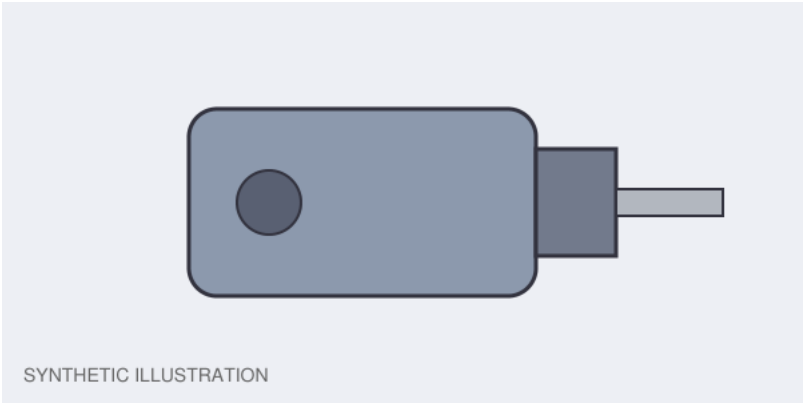

Figure 2 HX-200 pump, drive end (synthetic illustration)